MILITARY SPECIFICATION

SUPPORT SYSTEM, SPECIAL DIVER'S AIR (SDASS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the special diver's air support system hereinafter called the SDASS.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2}.

SPECIFICATIONS

FEDERAL

BB-A-1034	- Compressed Air, Breathing.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Method Of.
MIL-T-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4220
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STANDARDS

FEDERAL

FED-STD-101	- Test Procedures for Packaging Materials.
FED-STD-595	- Colors Used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-278	- Welding and Casting, Standard of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government documents, drawings, and Duplications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13229E4550	- Special Divers Air Support System (SDASS).

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: US ARMY BELVOIR RDE CTR, ATTN SATBE FMT, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

DEPARTMENT OF LABOR (DOL)

OSHA 29 CFR 1910 Occupational Safety and Health Standards.

(Application for copies should be addressed to: SUPT OF DCMNTS, GVT PRINTG OFC, WASHINGTON DC 20402.)

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 171-179	- Hazardous Materials Regulations.
DOT-E-8162	- Exemption to DOT FRP-l; Basic Requirement for FRP Type 3FC Cylinders.

(Application for copies should be addressed to: SUPT OF DCMNTS, GVT PRINTG OFC, WASHINGTON DC 20402.)

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U.S.NAVY/U.S. ARMY PUBLICATIONS

NCSC5110-AR-001 - Speclfication for Constructing Special Diver's Air Support System (SDASS).

(Application for copies should be addressed to: CSS, CODE 2510, PANAMA CITY FL 32407.)

FM 20-11-1 - Army Diving Manual Volume 1, Stk. No. 008-046-00110-3.

(Application for copies should be addressed to: SUPT OF DCMNTS, GVT PRINTG OFC, WASHINGTON DC 20402.)

2.2 Non-Government Duplications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. unless otherwise specified, the issues of documents not listed in the DoOISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazil Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to: AMERCN SCTY OF MECHL ENGRS, 345-E 47TH STRET, NEW YORK NY 10017.)

AMERICAN MELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Steel.

(Application for copies should be addressed to: AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The SDASS supports a one day, extended lightweight diving operation, or the secondary life support air source requirements for recompression chambers. The SDASS provides breathing air to diver personnel by accepting air from any of three separate air sources and properly distributing the air to one, two, or three divers at a maximum of 250 feet of sea water (FSW). The system

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provides constant depth monitoring of the divers by topside personnel. The SDASS piping, fittings, valves, gauges, -regulators, connecting high pressure (HP) and low pressure (LP) hoses, HP flasks and LP tanks shall be designed for working HP pressure of not less than 5,000 psig and LP pressure of not less than 350 psig. They shall be of materials as specified in the drawing package: TA13229E4550. Safety devices shall be provided as specified in the drawing package to protect the HP and LP compressed air system. The SDASS shall be capable of withstanding a hydrostatic pressure test of not less than five-thirds of the working pressure without leaking or damage to the SDASS. The SDASS shall be delivered in a clean state as specified in 3.12.2 when tested in accordance with 4.5.2.4. The SDASS shall be constructed in accordance with NCSC5110-AR-001 and all applicable drawings. The SDASS consists of the following major components:

a. Diver air control console (DACCI. The DACC is a man portable console which shall be installed on top of the volume tank assembly (VTA) during diving operations. The DACC shall receive HP air (5,000 psig) from the diver's air storage system (DASS) or LP air (250 psig) from a 88.5 cfm compressor. HP air shall enter the DACC through either the primary or secondary circuit, each consisting of HP shutoff valves, manual adjustable regulator, HP and LP gauges with isolation valves, and an emergency crossover valve. HP air shall be reduced to LP air and supplied to the VTA through one of two LP outlet valves. The LP circuit shall consist of a shutoff valve to control air from the 88.5 cfm compressor, LP gauge with an isolation valve, and an LP outlet valve. A bleed off valve shall be supplied for draining air from the DACC. The DACC shall also have three pneumo-fathometers for constant monitoring of the diver's depth. The DACC shall be connected to the VTA through a pair of flexible deck hoses.
b. Volume tank assembly (VTA). The volume tank assembly shall be a man portable, frame-mounted LP air reservoir with an eight-cubic foot, floodable volume. Low pressure air shall be received from the DACC through a pair of flexible deck hoses. The incoming LP air shall enter through the VTA manifold and pass through a check valve into the LP volume tank. Two flex whips shall deliver LP air from the LP tank the flip up control panel. The control panel shall have three diver LP air supply valves, a bleed off valve, a gauge, and gauge isolation valve. Diver's air shall leave the VTA and enter the diver's umbilical at the panel's "deep-sea" bulkhead fitting.
c. Diver air storage system (DASS). The diver air storage system (DASS) shall serve as a portable storage bank for high pressure air (5,000 psig). Each DASS rack shall contain two 1075-cubic-foot (3.15 cubic-foot floodable volume each) Kevlar-wrapped flasks housed in an aluminum stackable pipe frame. Each assembly shall consist of four racks. Each end of the rack shall have a separate control panel with an Wet and an outlet globe valve. In addition, the charging ends panel shall have a bleed off valve for draining air. The outlet end's panel shall have gauge and a gauge isolation valve. Four racks shall be interconnected with the DACC for dive operations, while four additional racks are interconnected with the chamber reducing station (CRS) to support recompression chamber operations. Each assembly of four racks shall be stacked two high and two abreast. Interconnections between individually stacked racks shall be made outlet to inlet with eight-foot-long HP deck hose.
d. Chamber reducing station (CRS). The CRS shall be an air monitoring and control assembly for the air flow to a recompression chamber. The CRS shall be a four man portable tubular frame enclosing a control panel and associated hardware with one HP (5,000 psig) and two LP (250 psig) inlets on the left

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side, and LP (40 to 350 psig) outlets on the right side. The control panel shall consist of primary and secondary HP shutoff valves, a manual adjust regulator, two LP inlet stop check valves, HP and LP gauges with isolation valves, LP outlet valve, and bleed off valve. The CRS shall receive HP air from the DASS and shall *receive* LP air from two 88.5 cfm compressors. Primary HP control circuit shall consist of a shutoff valve and a manual adjust regulator to control a dome (loader) regulator's discharge (40 to 350 lbs) and shall be capable of pressuring a 201 cubic-foot recompression chamber (inner lock) to 165 FSW (73.4 psi) in under 3 minutes. Secondary HP control circuit shall consist of a shutoff valve and a manual adjust regulator and continues recompression which allows at least 71 SCFM for 60 minutes at 5 atmosphere (ATMs) at a reduced flow rate. The LP control circuit shall consist of two valves to control air from one *or* two air compressors and shall be capable of pressuring the inner lock of a 201 cubic-foot recompression chamber in under 3 minutes. All air supplies shall feed a manifold and shall be controlled by an LP outlet valve to bulkhead fittings. The CRS shall be connected to the recompression chamber by two 10-feet long, LP hoses.

e. Deck and interconnecting hoses. HP deck hoses shall be used to connect the HP compressor to the DASS, the DASS to the DACC or CRS. HP hoses shall also be used to interconnect flask racks. LP deck hoses shall be used to connect the LP compressors to the DACC or CRS and the DACC to the VTA. LP deck hoses shall also be used to connect the CRS to the recompression chamber.

3.1.1 Drawings The drawings forming a part of this specification are end product drawings." No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Because of the life support nature of the SDASS, no deviations from the cleaning instructions found in the drawing package is permissible without the prior written approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the equipment. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) and in accordance with 4.3.

3.3 Material. Material shall be as specified herein and in the drawing package. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and drawing(s) package.

3.3.1 Material deterioration Prevention and control. The SDASS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the SDASS may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion (e.g., when fabricating the stainless steel LP volume tank. Lower grade steel or ferritic

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metal shall not be used as a tank support or as welding material). Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the SDASS may be newly fabricated from recovered materials to the maximum extent practicable, provided the SDASS produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the SDASS.

3.3.4 Cadmium plate. Cadmium plating and cadmium plated items shall not be used.

3.4 Transportability. The SDASS shall be capable of withstanding the shock, rigors, and vibrational stress encountered during air (both fixed wing and helicopter), sea, land, and rail transportation.

3.5 Watertightness. The SDASS shall be watertight when configured for operation, transportation, and storage.

3.6 Safety. The SDASS shall be designed and fabricated to prevent injury to operator and maintenance personnel. Nonfunctional rough or sharp edges, projecting points, or excessive lengths of fasteners shall be avoided. Extreme caution shall be taken to design, fabricated, and clean all components that will be exposed to high and low pressure compressed air. The SDASS shall conform to the following safety requirements:

a. Comply with the Hazardous Materials Regulations of the DOT 49 CFR 177, including placards and DOT-E-8162 exemption to DOT FRP-l; Basic requirement for FRP type 3FC cylinders.
b. Comply with DOL OSHA Standards (29 CFR 1910).
c. Eliminate nonfunctional sharp edges, projecting points, and excessive lengths of fasteners.

3.7 Human factors engineering. The SDASS shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472.

3.8 Component mounting and tiedown. The contractor shall be responsible for installation of the necessary mounting and tiedown devices to properly anchor the HP flex whips and LP deck hoses and to follow the applicable drawings when fabricating major component tiedown and anchor points to prevent movement during transit. Sensitive items such as gauges and valves shall be protected against the rigors of shock and vibration during movement.

3.9 Identification marking and Plates.

3.9.1 Identification plates. SDASS equipment shall be identified in accordance with MIL-STD-130, MIL-P-15024, and all applicable drawings.

3.9.2 Instruction plates. The identification of appropriate components (see 6.2) and the components proper location shall be marked by plates conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material.

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3.10 Treatment and painting. The portions of the equipment normally panted, shall be cleaned, treated and painted in accordance with the applicable drawings.

3.11 Government-1oaned property. When specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.4):

Item No.	Description	identification	Quantity
1	20 cfm, 5,000 pslg HP diving compressor	4310-01-291-8028	2 ea
2	88.5 cfm, 250 psig LP diving compressor	4310-01-1J3-8271	2 ea

3.12 Workmanship The equipment shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation or serviceability of the equipment. All reports and documentation required shall be legible, free of smudges, stains, or foreign material that prevents the document(s) from being clearly understood.

3.12.1 Fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury or fatigue to the metal. Mechanical process flame cutting of steel may be employed instead of shearing or sawing. Burned surfaces of flame-cut steel shall be ground or machined to remove ash and cooling checks. All burrs and rough edges shall be removed. Overheating of the metal shall be avoided. All bends of a major character shall be made with metal dies or fixtures to ensure uniformity of size and shape.

3.12.2 Cleaning. All surfaces exposed to high and low *pressure* compressed air, including valves, plumbing, dome loaders, regulators, gauges and hoses shall be cleaned and documented in accordance with the drawing package (see 6.7).

3.12.3 Riveted connections. Rivets holes shall be accurately formed and shall have the burrs and sharp edges removed. Rivets shall completely fill the rivet holes. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the members,

3.12.4 Welders and welding operators. Welders and welding operators assigned to work covered by this specification and drawing package shall be qualified in accordance with ASME Boiler and Pressure Vessel Code, Section IX, AWS D1.1, or MIL-STD-278. The contractor is responsible for determining that automatic welding operators are capable of producing quality welds in accordance with the above codes (see 6.9).

3.12.5 Welding. Welding procedures and welds shall be in accordance with ASME, AWS, and MIL-STD-278 codes. The surfaces of parts to be welded shall be free from rust, scale, paint, grease (both organic and inorganic) and other forms of foreign matter that will cause an inferior weld. Welds shall transmit stress without permanent deformation or failure when parts connected by the welds are as specified herein.

3.13 Environmental requirements.

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3.13.1 Operational temperature. The SDASS shall be operational in any ambient air temperature between 0 and +120 °F.

3.13.2 Storage temperature. The SDASS unit shall not be damaged by storage in any ambient temperature from -25 to +155 °F.

3.13.3 Precipitation. The SDASS shall not be damaged or become inoperative when exposed to rain ranging from a drizzle to a heavy rain of 1 quart per minute per foot of wetted surface impinging upon the components of the SDASS with a constant 30 mph wind.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as othewise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 ResponsibilitY for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components *and* materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Parts and components. Parts and components detailed on the drawing shall be inspected in accordance with the quality assurance provisions (QAP) (see 6.5.1) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection.

4,2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Comparison inspection (see 4.6)
c. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. SDASS equipment shall be examined for the defects listed in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the first article shall be subjected to the tests marked "X" in column 1 of table 1. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for defects listed in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 2 of table 1. Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The SDASS shall be examined for the following defects:

101. Components and equipment missing or not as specified (see 3.1).
102. Material(s) not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the compressor (see 3.3.3).
106. Cadmium plating used (see 3.3.4).
107. Safety not as specified (see 3.6).
108. Human factors engineering not as specified (see 3.7).
109. Component anchorage or storage not as specified (see 3.8).
110. Identification marking not as specified (see 3.9.1).
111. Instruction plates not as specified (see 3.9.2).
112. Treatment and painting not as specified (see 3.10).
113. Workmanship not as specified (see 3.12).
114. Fabrication not as specified (see 3.12.1).
115. Cleaning procedure and cleaning not as specified (see 3.12.2 and applicable drawings).
116. Riveted connections not *as* specified (see 3.12.3).
117. Welders and welding not as specified (see 3.12.4 and 3.12.5).
118. Welders' certification not available, not as specified (see 3.12.4).
119. Any part or component not in accordance with the requirements as shown on the drawings.
120. Any dimensions, other than those identified by the QAP requirements, shown on the drawings, not as specified.
121. Assembly not complete or not correct (see applicable drawing).
122. Rivets or other fasteners missing (see applicable drawing).
123. Parts *or* components misidentified or dislocated (see applicable drawing).

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4.5.2 Tests.

4.5.2.1 Test schedule. Tests shall be in accordance with table I.

TABLE I. Test schedule.

First article	Quality conformance	Test	Requirement paragraph	Test paragraph
1	2	3	4	5
X	X	Compressed air cleanliness	3.12.2	4.5.2.4
X	X	Pressure and leak	3.1	4.5.2.5
X	-	Watertightness (precipitation)	3.5	4.5.2.6
X	-	Flow test	3.1 & 3.12.2	4.5.2.7
X		Shock and vibration (pendulum - impact)	3.4 & 3.8	4.5.2.8
X		High and low temperature	3.13.1 & 3.13.2	4.5.2.9

4.5.2.2 Test schedule and conditions. The order of the tests for the first article and production models, when applicable, are as follows: Shock and vibration (pendulum - impact); component tiedown (anchoring); watertightness (precipitation), high and low temperature; pressure and leak test; compressed air flow; and compressed air cleanliness. The compressed air analysis sampling done at the end of the compressed air cleanliness procedure can be done after the compressed air flow procedure in order to preclude charging of the SDASS twice. . The tests shall be performed at ambient temperature at the place of testing as appropriate, unless otherwise specified in this specification. Unless specified herein, tests shall be conducted with all SDASS components and accessories installed and functioning as specified.

4.5.2.3 Test log. A first article test log covering all tests in table I shall be established tabulating the appropriate test data. A quality conformance test log covering all tests required in table I shall be established test log covering all tests required in table I shall be established tabulating the appropriate test data.

4.5.2.4 Compressed air cleanliness. The compressed air cleanliness sampling and analysis shall be conducted in accordance with the procedures and requirements as stated in the drawing package and BB-A-1034 constituent level for table I (pressured container air), grade A (see 6.7).

4.5.2.4.1 Failure criteria. Nonconformance to the drawing package and BB-A-1034 procedures and results shall constitute failure of this test.

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4.5.2.5 Pressure and leak test. The SDASS unit shall be charged to working pressure for each component not less than 5,000 psig for the HP items and 240 psig for the LP items. The SDASS outlet ports to the diver umbilical on the VTA and the outlet port on the CRS to the recompression chamber shall be plugged leak tight. The Inlet port on the DASS to the 20 CFM compressor shall remain connected to the compressor and shall be secured leak tight. The SDASS unit shall remain charged with HP and LP air for a l-hour period; the contractor shall apply a "non-ionic" soap and water "wash" to all valves, gauges, plumbing, regulators, dome loaders, hoses, fittings, and welds to Identify potential leaks. The l non-ionic soap and water wash" shall be applied every 15 minutes. The contractor will be allowed to tighten any valve fitting, gauge fitting, dome loader fitting, hose fitting, and pipe fitting to stop any leaks that may be occurring during this l-hour period. The contractor shall not be allowed to change out any faulty component or to repair any welds during this period. Upon completion of the l-hour period the SDASS shall hold its working pressure for a 24-hour period for high pressure and 12-hour period for low pressure. The contractor shall be allowed to top off the high and low pressure items to the high pressure working limit (5,000 psig) and the low pressure working limit (250 psig). The 12-hour period shall be run simultaneously with the 24-hour high pressure period. The contractor shall record the beginning high and low gauge pressure and the ending high and low gauge pressure.

4.5.2.5.1 Failure criteria. Nonconformance to the test procedure; high pressure drop of more than 1 percent of the starting high pressure over the 24-hour period; low pressure drop of more than 5 percent of the starting low pressure over the 12-hour period; any leaks in welds; any leaks in the piping, valve, gauges, regulators, dome loaders, or fittings that cannot be corrected during the first l-hour leak check period prior to the 24-hour pressure test. The contractor will be given the opportunity to determine the cause of any leakage then repair those leaks to the SDASS system that would not be initially corrected after the 24-hour pressure test period is completed. The contractor shall then rerun the entire pressure test until all of the objectives of the pressure test are met.

4.5.2.6 Watertightness (precipitation). The contractor shall place the SDASS, in its operational state and storage/transportation configuration state, on a horizontal surface and expose the unit to natural or simulated rain. The test shall be conducted twice, once in the SDASS'S operational state and then again in *its* storage/transportation state. The volume of rain impinging on the horizontal surface shall be not less than 1 quart per minute per square foot. Simulated rain shall originate from a source which produces droplets uniform distribution and velocity similar to natural rain at a point of contact with the SDASS. Expose the top, all four sides, and ends to the wind-blown precipitation. After the rain exposure, operate the SDASS in accordance with FM-20-11-l for a 3-man, lightweight and deep sea diving operation. The simulated depth for each dive shall be 250 FSW. The contractor can simulate the diving depths by using a pressure controlled regulator(s) on the outlet side of the VTA along with a flow meter to record low pressure air flow to the "diver" on the outlet side of the VTA.

4.5.2.6.1 Failure criteria. Water leakage in weather prepared areas, malfunction of the SDASS; damage to the SDASS caused by water, and noncompliance to the flow rates expected for simulated lightweight and deep sea dives due to water shall constitute failure of this test.

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4.5.2.7 . Flow test The SDASS shall be subjected to a total of 5 each simulated lightweight dive missions for three divers down to a depth of 250 FSW, 5 each 5 each simulated recompression chamber runs to the 165 FSW within 3 minutes for each run for both the inner and outer lock of s standard 201 cubic foot Army/Navy recompression chamber. Air flow requirements are identified in table 6-2 of FM 20-11-1. A lightweight diving set mission requires 12.01 actual cubic foot per minute (ACFM) per diver at one atmosphere to equal 1.4 ACFM at 250 FSW; a deep sea diving set mission requires 38.61 ACFM per diver at one atmosphere to equal 4.5 ACFM at 250 FSW; "descent" to 165 FSW for the recompression chamber shall be as fast as possible but no slower than 3 minutes from the time the chamber leaves the surface until a depth of 165 FSW is attained. The contractor can use flow meters and pressure regulators to simulate the measure depths of 250 FSW for the diving operations and to simulate depths of 165 FSW for the recompression chamber operation.

4.5.2.7.1 Failure criteria. Nonconformance to flow rates as identified in 4.5.2.7 any SDASS hardware failure that restricts the flow in accordance to required flow rates, and failure to pressurize the recompression chamber in accordance with "descent" time of 3 minutes or less shall constitute a failure of this test.

4.5.2.8 Shock and vibration (pendulum - impact). Each separate SDASS component with the exception of the DACC will be subjected to the pendulum - impact test in accordance with FED-STD-101, method 5012, except that the vertical height of drop shall be 6 inches. Upon completion of the test the SDASS (complete system) shall be operated and tested as specified in 4.5.2.5 for 30 minutest.

4.5.2.8.1 Failure criteria. Any cracks, leaks, breaks to the SDASS frames, plumbing, malfunctions to gauges, valves, regulators, and dome loaders, deformity to the SDASS components, or failure to follow test procedures of FED-STD-101, method 5012, shall constitute failure of this test.

4.5.2.9 Low and high temperature test. The SDASS shall be subjected to a high temperature of 110 ±2 °F for 12 hours or until the high and low pressure readings have stabilized for at least 60 minutes at the test temperature at which time the SDASS shall be "topped off" so all high pressure gauge readings are at 5,000 psig and low pressure gauge readings are 250 psig, whichever occurs first. The SDASS shall then be tested as specified in 4.5.2.5 and 4.5.2.7. The SDASS shall be subjected to a low temperature of O ±2 °F. The SDASS shall then be tested as specified in 4.5.2.5 and 4.5.2.7 using the same "time" criteria as stated for high temperature testing. Both high and low temperature tests must be conducted separately from testing already conducted.

4.5.2.9.1 Failure criteria. Failure to follow procedures as established in 4.5.2.5 and 4.5.2.7, any hardware failure that restricts the flow of compressed air in accordance to established flow rates, any failure of hardware that would cause the diving operation or recompression chamber operation to be aborted, any low temperature failure where ice is the cause of restricting flow or malfunction to the hardware that would cause a diving operation or recompression chamber operation to be aborted if the moisture content of the compressed air is within acceptable limits of BB-A-1034 shall constitute a failure of this test.

4.6 Comparison inspection. The Government may selected at any time any production item during the contract period and subject that equipment to the

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examination specified in 4.5.1 and all tests specified in 4.5.2 *to* determine conformance to the requirements of the specification and associated drawings. The inspection will be performed by the Government, at a site selected by the Government, on SDASS systems or components selected at random from those which have bene accepted by the Government and will not include the previously inspected first article model. In addition to any tests so specified as part of the inspection, the Government reserves the right to conduct any and all other tests contained In this specification as part of the Government quality assurance inspection. Failure of such additional tests shall have the same effect as failure of those tests as identified in this specification if done by the contractor. This inspection shall be identified as a "comparison inspection" by the Government.

4.6.1 Inspection failure. Failure of a comparison inspection equipment to meet any requirements specified herein during and as a result of the examination and tests in 4.6 shall be cause for rejection of the inspection equipment and shall also be cause for refusal by the Government to continue acceptance of production equipment until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on equipment previously accepted and produced under the contract. Any deficiencies found as a result of the inspection will be considered prima facie evidence that all equipment accepted prior to the completion of inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the Government contracting officer.

4.7 Inspection of packaging.

4.7.1 First article pack inspection.

4.7.1.1 Examination. Examine the first article pack for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Test. The first article pack of a complete SDASS for levels A and B, shall be subjected to the rail-car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking and bracing, or any visual damage to the components shall constitute failure of the first article pack and shall be cause for rejection.

4.7.2 OualitY conformance inspection of pack.

4.7.2.] Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit product.

4.7.2.2 Sampling, Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

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4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

124.	The DACC not preserved in accordance with the correct method of the referenced document for level A or C (see 5.2.1.1 and 5.2.2).
125.	The CRS not preserved as specified for level A or C (see 5.2.1.2 and 5.2.2).
126.	The VTA and DASS inlets/outlets not covered with caps or plugs as specified for level A or C (see 5.2.1.3 and 5.2.2).
127.	The hoses not coiled and secured as specified and the number of hoses exceed the weight limitation of the boxes as specified for level A or C (see 5.2.1.4 and 5.2.2).
128.	Technical publications not preserved as specified by either method of the referenced document for level A or C (see 5.2.1.5 and 5.2.2).
129.	The DACC and technical publications not packed together as specified for level A or B (see 5.3.1.1 and 5.3.2).
130.	Blocking and bracing not as specified for level A or B (see 5.3.1.1 and 5.3.2).
131.	Box closure and strapping not as specified by the box specification for level A or B (see 5.3.1.1 and 5.3.2).
132.	Strapping not zinc-coated as specified for level A (see 5.3.1.1).
133.	The CRS, VTA and DASS not packed as specified for level A or B (see 5.3.1.2, 5.3.1.3, and 5.3.2).
134.	Deck and interconnecting hoses not packed together as specified for level A or B (see 5.3.1.4 and 5.3.2).
135.	Level C packing not as specified (see 5.3.3).
136.	Marking incorrect, illegible, or missing for level A, B, or C (see 5.4).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.2). The first article pack may be accomplished utilizing either first article or production equipment. If a first article model is utilized, any preservation and packaging shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the first article and production equipment.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Diver air control console. Each DACC shall be preserved in accordance with method IIc of MIL-P-116 and then individually placed in a close-fitting box conforming to PPP-B-636, class weather-resistant.

5.2.1.2 Chamber reducing station. Each CRS shall be preserved as specified in 5.2.1.1.

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5.2.1.3 Volume tank assembly and diver air storage system. Any inlets/outlets on the VTA or the DASS shall be covered with caps or plugs.

5.2.1.4 Deck and interconnecting hoses. The hoses shall be loosely coiled and secured to a diameter than will not cause damage or kinks. The hoses shall then be placed in boxes conforming to PPP-B-636, weather-resistant, in quantities not to exceed the weight limitations of the box. Only like hoses shall be placed in a box together.

5.2.1.5 Technical publications. Any technical publications shall be preserved in accordance with method IC-1 or IC-3 of MIL-P-116.

5.2.2 level C. Each component of the SDASS shall be preserved as specified for level A except the DACC and CRS shall be preserved in accordance with method IA-8 of MIL-P-116 in lieu of method IIc and any boxes shall be domestic class.

5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Diver air control and technical publications. Each DACC and any technical publications, preserved as specified in 5.2, shall be packed together in a container conforming to PPP-B-601, overseas type, grade A, style optional. Blocking and bracing shall be in accordance with MIL-STD-1186 to prevent movement and damage. Box closure and strapping shall be in accordance with the box specification except the strapping shall be zinc-coated.

5.3.1.2 Chamber reducing station. Each CRS, preserved as specified in 5.2, shall be packed as specified in 5.3.1.1.

5.3.1.3 Volume tank assembly and diver air storage system. Each VTA and DASS, preserved as specified in 5.2, shall be packed as specified in 5.3.1.1.

5.3.1.4 Deck and interconnecting hoses. All hoses, preserved as specified in 5.2, shall be packed together in a box conforming to PPP-B-601, overseas type, grade A, style optional.

5.3.2 Level B. The components of the SDASS, preserved as specified in 5.2, shall be packed as specified for level A except the boxes shall be domestic type and the strapping may be zinc-coated.

5.3.3 Level C. The components of the SDASS, preserved as specified in 5.2, shall be packed in such a manner to afford protection against damage and deterioration during any brief storage periods, handling and shipment from the contractor to the initial destination.

5.4 Marking. Unless otherwise specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The SDASS is intended to be a high pressure compressed air storage system, to reduce the stored high pressure compressed air; distribute and control the low pressure air to up to three divers to depths of 250 FSW using either the lightweight diving equipment set (100 foot) or the deep sea diving equipment set (120 feet) and; to distribute the low pressure compressed air to a double lock aluminum or steel hull, 2-man recompression chamber. The SDASS will be used with 2 low pressure 88.5 cfm diving air compressors and a high pressure 20 cfm diving air compressor.

6.2 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article model (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
d. Identification of components and location of plates required (see 3.9.2).
e. When the property listed in 3.11 will be loaned by the Government.
f. Time frame required for submission of the first article pack (see 5.1).
g* Level of preservation and packing required (see 5.2 and 5.3).
h. Any special marking (see 5.4).

6.3 Data requirements. When this specification is used for an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

	Paragraph No.	Data requirements	Applicable DID No.
a.	4.5.2.3	First article inspection procedure	DI-T-4901
b.	4.5.2.3	First article inspection report	DI-T-4902

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-1, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the

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requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.11.

6.6 Definitions.

6.6.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicate the minimum requirements which may be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6.2 Displacement. Displacement is defined as dislodgement of any SDASS - DACC, DASS, VTA, or CRS high and low pressure fittings, hoses, gauges, valves, or plumbing from its designated position. Movement of these items within its designated position without dislodgement or damage is permitted.

6.7 First article pack. Any changes *or* deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor's obligation to preserve, package, pack, and mark the equipment in accordance with this specification and associated drawings.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair/spare parts, maintenance cards in accordance with U.S. Navy planned maintenance system (3M), instructions on shipment of equipment instructional manuals to be furnished for each time of equipment. A suggested paragraph is as follows:

> "Shipment of SDASS shall include repair/spare parts, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.9 Welders and welding operations. The contracting officer should consider requalifying welders or welding operations in the event of evidence of poor welds (see 3.12.4).

6.10 Subject term [key word) listing.

Deep sea diving set (120-foot)
High pressure compressed air
Lightweight diving set (100-foot)
Low pressure compressed air
Recompression chamber, two-man

Custodians:
Army - ME
Navy - SH

Preparing activity:
Army - ME

Project 4220-0382

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
   The submitter of this form must complete blocks 4, 5, 6, and 7.
2. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-53120	2. DOCUMENT DATE (YYMMDD) 940125

3. DOCUMENT TITLE
SUPPORT SYSTEM, SPECIAL DIVER'S AIR (SDASS)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
c. ADDRESS (Include Zip Code) S ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340	

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